Determinant

Mind map

Order and expansion

- $ad - bc$; order 3 along R_1
- Expand along most zeros
- Sarrus: order 3 only

Minor and cofactor

- $C_{ij} = (-1)^{i+j} M_{ij}$
- Own row $= \Delta$; other $= 0$
- $|C_{ij}| = \Delta^{n-1}$

Properties P1-P6

- $D' = D$; swap $\to -D$
- Proportional rows $\to 0$
- k^n; split sum; factor

Transformations

- $R_i + \lambda R_j$: unchanged
- $xR_i + \ldots$: $\times x$
- Several operations: in order

Standard determinants

- $(a-b)(b-c)(c-a)$
- $\times(a+b+c)$, $\times(ab+bc+ca)$
- Cyclic: $-(a^3+b^3+c^3-3abc)$

Two operations; AP

- Row of 1's: subtract
- Add rows: common factor
- AP: $C_1 - 2C_2 + C_3 = 0$

Multiplication

- $|A||B| = |AB|$, 4 ways
- $|C_{ij}| = D^2$
- Spot sums of products

Equations in x

- Simplify, factor, roots
- Constant term $\Delta(0)$
- Trig: reduce to $a + b\sin 2x$

System: two variables

- Ratio test: unique/infinite/none
- Lines: meet/coincide/parallel
- Three lines: $D = 0$ concurrent

Cramer's rule

- $x = D_x/D$, ...
- Replace column by constants
- Needs $D \neq 0$

Three variables: cases

- $D \neq 0$: unique
- $D = 0$, $D_i \neq 0$: none
- All zero: check; $R_3 = \lambda R_1 + \mu R_2$

Planes in 3-D

- Point / line / none
- Prism: no common point
- Three parallel: exception

Homogeneous system

- Always trivial solution
- Non-trivial iff $D = 0$
- Planes through origin